Operational Audit Case Study

PHARx Logistics
Operational Audit

A data-led audit of inventory performance at Park Royal Depot.

COMMISSIONED BY

Head of Logistics & Corporate Finance

METHODOLOGY

Google BigQuery SQL Audit

ANALYSIS PERIOD

12-Month Transaction History

AGENDA

SESSION OUTLINE & EXPECTATIONS

EXECUTIVE SUMMARY

OPERATIONAL AUDIT OVERVIEW

SITUATION

01 . Depot Bottleneck

High-cost specialty medicines are expiring on shelves, while standard daily prescriptions are consistently suffering stockouts.

Heavy financial loss

MISSION

02 . Strategic Task

Audit 12 months of sales and inventory logs to align supply with actual demand and improve warehouse efficiency.

Map demand & supply velocities

RESULTS

03 . Core Objectives

- ✓ **Financial Efficiency:** Minimize capital loss from overstock.
- ✓ **Supply Reliability:** Achieve zero-stockout operations.

Balanced capital with zero stockouts

OPERATIONAL HYPOTHESIS AUDIT

Hypothesis 1: Specialty Capital Drag

Specialty medications drive 80% of waste due to infrequent bulk orders that ignore actual turnover and expiry risks.

Hypothesis 2: Daily Stockout Instability

Everyday prescriptions face stock outs because of rigid, fixed-interval ordering fails to adapt to demand surges or lead-time delays.

DATA STORAGE, STRUCTURE & ROCCC AUDIT

Relational BigQuery Schema

```
        [ fact_sales ]                      [ fact_inventory_logs ]
       ────────────────                    ────────────────────────
        transaction_id (PK)                  log_id (PK)
        sales_date                           log_date
  ┌───> product_id (FK)                ┌───> product_id (FK)
  │     quantity_sold                        current_stock_on_hand
  │     total_revenue                        expiry_date
  │                                          supplier_lead_time_days
  │
  │             [ dim_products ]
  │            ──────────────────
  └──────────   product_id (PK)
                product_name
                drug_category
                unit_cost
                unit_price
```

ROCCC Framework Audit

- **Reliable:** Mirrors real distributor sales patterns.
- **Original:** Uses relational warehouse database schemas.
- **Comprehensive:** Audits all products and historical logs.

Privacy, Compliance & Safety

- **Zero Breach Risk:** Built on engineered data, eliminating privacy exposure.
- **Enterprise Governance:** Enforced data protection using isolated cloud roles in Google BigQuery.

STRATEGIC CORE OBJECTIVES

CONSOLIDATED LOGISTICS STRATEGY

A unified analytical approach to logistics optimisation, addressing both high-cost specialty waste and everyday high-volume stockouts as a single interconnected inventory flow.

01 . Demand Profile Alignment

Established baseline demand by calculating average daily sales and aligning bulk orders directly with each product's demand.

✦ **Eliminating over- & understocking**

02 . Expiration Timeline Syncing

Prevented high-cost drug wastage by dynamically syncing active inventory volumes directly with manufacturing expiration dates.

✦ **Optimising shelf-life coverage**

03 . Lead Times & Safety Stock Buffers

Supplier Delivery Timelines

Factored true supplier delivery timelines directly into the loop to eliminate coverage gaps.

Safety Stock Buffers

Established precise, item-level buffers to guarantee zero stockouts on critical daily products.

✦ **Securing supply chain resilience**

REPLENISHMENT MODELING

THE OPTIMISATION **MODEL**

FREQUENCY WINDOW

Calculating Optimal Ordering Cycle

place_order_every_x_days: Defines a safe, automated buffer window to prevent capital stagnation while avoiding premature reorders.

```
-- 01 | Calculating Optimal Ordering Cycle
CAST(
  (average_inventory_in_stock / avg_units_sold_per_day)
  - average_supplier_lead_time_days - 1
AS INT64) AS place_order_every_x_days
```

**Note: Ensures stock arrives just before the safety buffer depletes*

ORDER QUANTITY

Calculating Optimal Restock Volume

order_quantity: Establishes the precise volume boundary needed to maintain optimum shelf levels without introducing stockout risk.

```
-- 02 | Calculating Optimal Restock Volume
CAST(
  (place_order_every_x_days * avg_units_sold_per_day)
  + (average_supplier_lead_time_days * avg_units_sold_per_day)
AS INT64) AS order_quantity
```

**Note: Synchronises incoming shipment volumes with actual sales velocities*

Drug category	Product ID	Product Name	Place order every (x) days	Order quantity
Cardiology	SKU_1014	Amlodipine	49	226
General Antibiotics	SKU_1048	Levofloxacin	45	221
General Antibiotics	SKU_1041	Amoxicillin	33	163
Diabetology	SKU_1010	Canagliflozin	30	192
Diabetology	SKU_1001	Insulin Glargine	9	84

AUDIT IMPACT

CURRENT INVENTORY VS. OPTIMISED IMPACT

A comparison of baseline inventory holding value against the optimised replenishment model.

Total inventory savings

£372,636 .13 saved

✦ 6.0% Capital Efficiency Improvement

£6,240,881.18 Baseline Inventory Value (Before Audit)

£5,868,245.05 Optimised inventory value (Post-Audit Model)

CATEGORY BREAKDOWN

Most of all capital saved derived from specialty drug categories.

Oncology Saved £190,000

Current: **£3.19M** | Optimised: **£3.00M**

Immunology Saved £180,000

Current: **£2.87M** | Optimised: **£2.69M**

Other Categories Saved £2,636

Cardiology, Diabetology, & General Antibiotics combined

Current: **£0.19M** | Optimised: **£0.17M**

STRATEGIC ACTION & LOGISTICS ROADMAP

1 Phase 1:
Continuous Execution

- Embed the replenishment model into standard operating procedure.
- Automatically adjust as market sales velocities and lead times fluctuate over time.

2 Phase 2:
Capital Reinvestment

- Reinvest the £372,636.13 directly into model automation and growth.

3 Phase 3:
Automate & Scale

- Turn the model into an auto-ordering tool.
- The model will place orders automatically based on real-time demand.
- Factors in product expiry, supplier lead time, and safety buffer.

***Automated Expansion Signaling:** When the automated model flags that optimal order volumes cannot be received due to a lack of physical shelf space, it serves as a data-backed trigger for warehouse expansion—directly funded by the capital saved from the initial model deployment.

OPERATIONAL AUDIT CONCLUDED

Questions?

Thank you for your time.

The audit models are configured, tested, and ready for deployment to secure working capital and eliminate stockout risks.

DATA SYSTEM
Google BigQuery

AUDITED FACILITY
Park Royal Depot, London

IMAGE SOURCES

https://static.vecteezy.com/system/resources/thumbnails/073/215/008/small/medication-drawer-partially-open-with-various-prescription-bottles-in-a-clinical-setting-featuring-cool-tones-and-a-modern-aesthetic-with-shallow-depth-of-field-photo.jpeg

Source: www.vecteezy.com